

SAT Unit 2.1.2

Equivalent Expressions

Jack Zeng

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Overview

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- 2 Expansion
- 3 Factoring
- 4 Fraction
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Combine Like Terms & Simplify

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Combining Like Terms

Definition: Combine terms with the exact same variables and powers.

Example:

$$2(2a^2 - 3a^2b^2 - 4b^2) - (a^2 + 5a^2b^2 - 10b^2)$$

Distribute:

$$= 4a^2 - 6a^2b^2 - 8b^2 - a^2 - 5a^2b^2 + 10b^2$$

Combine like terms:

$$= 3a^2 - 11a^2b^2 + 2b^2$$

Simplifying Fractional Expressions

Rules for Simplifying:

- Look for common factors in numerator and denominator.
- Apply factoring where possible.
- Cancel shared factors.

Example:

$$\frac{4xy(x+y)(x-y)}{6x(x-y)} = \frac{2y(x+y)}{3}$$

Note: Do not cancel terms unless they are factors!

Question: Combining Like Terms

Which expression is equivalent to

$$(3x^3 + 8x^2 - 4x) + (7x^2 - 11x - 7) ?$$

Options:

A. $3x^3 + x^2 - 15x - 7$

B. $3x^3 + 15x^2 - 15x - 7$

C. $10x^5 - 7x - 7$

D. $15x^4 + 3x^3 - 15x^2 - 7$

Answer Explanation: Combining Like Terms

Combine like terms:

$$(3x^3+8x^2-4x)+(7x^2-11x-7) = 3x^3+(8x^2+7x^2)+(-4x-11x)-7 = 3x^3+1$$

Correct Answer: B

Expansion

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Expansion Using FOIL

FOIL Method is a technique used to expand the product of two binomials.

FOIL stands for:

- **F** – First: Multiply the first terms in each binomial
- **O** – Outer: Multiply the outer terms
- **I** – Inner: Multiply the inner terms
- **L** – Last: Multiply the last terms in each binomial

Example: Expand $(x + 3)(x - 5)$

$$\text{First: } x \cdot x = x^2$$

$$\text{Outer: } x \cdot (-5) = -5x$$

$$\text{Inner: } 3 \cdot x = 3x$$

$$\text{Last: } 3 \cdot (-5) = -15$$

Combine like terms:

$$x^2 - 5x + 3x - 15 = x^2 - 2x - 15$$

Correct Answer:

$$x^2 - 2x - 15$$

Question: Find the Value of ab

The equation below is true for all x , where a and b are constants:

$$(ax + 3)(5x^2 - bx + 4) = 20x^3 - 9x^2 - 2x + 12$$

What is the value of ab ?

Options:

- A. 18
- B. 20
- C. 24
- D. 40

Answer Explanation: Find the Value of ab

We expand the left-hand side:

$$(ax + 3)(5x^2 - bx + 4) = 5ax^3 - abx^2 + 4ax + 15x^2 - 3bx + 12$$

Now combine like terms:

$$5ax^3 + (-ab + 15)x^2 + (4a - 3b)x + 12$$

Match with the right-hand side:

$$20x^3 - 9x^2 - 2x + 12$$

Equating coefficients:

$$5a = 20 \Rightarrow a = 4$$

$$-ab + 15 = -9 \Rightarrow -ab = -24 \Rightarrow ab = 24$$

Correct Answer: C

Question: Finding the Value of h

The expression $5x^5 + 6x^4 - 8x^3$ can be rewritten as

$$(x^3 - hx^2)(5x^2 + 10x),$$

where h is a constant. What is the value of h ?

Answer Explanation: Finding the Value of h

We are given:

$$(x^3 - hx^2)(5x^2 + 10x)$$

Use distributive property:

$$x^3(5x^2 + 10x) - hx^2(5x^2 + 10x) = 5x^5 + 10x^4 - 5hx^4 - 10hx^3$$

Combine like terms:

$$5x^5 + (10 - 5h)x^4 - 10hx^3$$

Compare with the original expression:

$$5x^5 + 6x^4 - 8x^3$$

Match coefficients:

$$10 - 5h = 6 \Rightarrow 5h = 4 \Rightarrow h = \frac{4}{5}$$

Correct Answer:

$$\boxed{\frac{4}{5}}$$

Factoring

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Factoring Basics

Factoring is the process of expressing a polynomial as a product of simpler expressions.

Step 1: Factor out the Greatest Common Factor (GCF)

$$6x^2 + 9x = 3x(2x + 3)$$

Step 2: Factoring Trinomials when $a = 1$

Method: Find two numbers that:

- Multiply to c
- Add to b

Example:

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

Tip: Always factor out the GCF before factoring the trinomial.

Factoring When $a \neq 1$ (Cross Method)

When the leading coefficient $a \neq 1$, we use the trial-and-error or Chinese Cross Method :

Example:

$$6x^2 + 7x + 2 \Rightarrow (3x + 2)(2x + 1)$$

Reasoning:

- $3x \cdot 1 = 3x$
- $2x \cdot 2 = 4x$
- $3x + 4x = 7x$

General Tip:

- Look for two pairs of factors of a and c whose cross-products add to b .
- Always check your result by expanding (FOIL method).

Question: Factoring a Quadratic Expression

Which of the following expressions is(are) a factor of

$$3x^2 + 20x - 63?$$

- I. $x - 9$
- II. $3x - 7$

Options:

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Answer Explanation: Factoring a Quadratic Expression

We are given the quadratic expression:

$$3x^2 + 20x - 63$$

Let's factor it:

$$3x^2 + 20x - 63 = (x - 3)(3x + 21)$$

But:

$$(x - 9)(3x - 7) = 3x^2 - 7x - 27x + 63 = 3x^2 - 34x + 63 \neq 3x^2 + 20x - 63$$

Let's check factoring by grouping:

$$3x^2 + 27x - 7x - 63 = 3x(x + 9) - 7(x + 9) = (x + 9)(3x - 7)$$

So the factors are:

$$(x + 9)(3x - 7)$$

Correct Answer:

B

Question: Finding the Smallest Possible Factor Constant

One of the factors of

$$2x^3 + 42x^2 + 208x$$

is $x + b$, where b is a positive constant.

What is the smallest possible value of b ?

Answer Explanation: Finding the Smallest Possible Factor Constant

We are given:

$$2x^3 + 42x^2 + 208x$$

First, factor out the GCF:

$$2x(x^2 + 21x + 104)$$

Now, factor the quadratic using cross multiplication (a.k.a. the AC method). We're looking for two numbers that multiply to:

$$1 \cdot 104 = 104 \quad \text{and add up to } 21.$$

The pair is 13 and 8, since:

$$13 \cdot 8 = 104 \quad \text{and} \quad 13 + 8 = 21.$$

So we factor the quadratic:

$$x^2 + 21x + 104 = (x + 13)(x + 8)$$

Therefore, the full factorization is:

Question: Factoring with Decimal Coefficients

The expression

$$0.36x^2 + 0.63x + 1.17$$

can be rewritten as

$$a(4x^2 + 7x + 13),$$

where a is a constant. What is the value of a ?

Answer Explanation: Factoring with Decimal Coefficients

We are given:

$$0.36x^2 + 0.63x + 1.17 = a(4x^2 + 7x + 13)$$

Match the coefficients:

$$a \cdot 4 = 0.36 \Rightarrow a = \frac{0.36}{4} = 0.09$$

Correct Answer:

Factoring Rational Expressions (Part 1)

Definition: A rational expression is a fraction where both the numerator and the denominator are polynomials.

Goal: Simplify by factoring both the numerator and denominator, then cancelling common factors.

Steps to Simplify:

- **Factor** the numerator and denominator completely.
- **Identify common factors** in both.
- **Cancel out** the common factors.
- **State restrictions:** Any value that makes the denominator zero must be excluded from the domain.

Note: Only factors (not terms) can be cancelled.

$$\frac{x^2 - 9}{x^2 - x - 6} = \frac{(x - 3)(x + 3)}{(x - 3)(x + 2)} = \frac{x + 3}{x + 2} \quad \text{with } x \neq 3, -2$$

Factoring Rational Expressions (Part 2)

Special Factoring Patterns to Know:

- Difference of squares: $a^2 - b^2 = (a - b)(a + b)$
- Perfect square trinomials: $a^2 \pm 2ab + b^2 = (a \pm b)^2$
- Difference and sum of cubes:

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

Example:

$$\frac{4x^2 - 9}{2x + 3} = \frac{(2x - 3)(2x + 3)}{2x + 3} = 2x - 3 \quad \text{with } x \neq -\frac{3}{2}$$

Tip:

- Always check if the polynomial is factorable before attempting to cancel.
- Watch for common mistakes like cancelling terms instead of factors.

Question: Polynomial Division

Given:

$$\frac{x^2 + 6x - 7}{x + 7} = ax + d$$

The equation is true for all $x \neq -7$, where a and d are integers.

What is the value of $a + d$?

- A. -6
- B. -1
- C. 0
- D. 1

Answer Explanation: Polynomial Division

We are dividing a quadratic by a linear binomial:

$$\frac{x^2 + 6x - 7}{x + 7}$$

We use polynomial division or factoring. First, factor the numerator:

$$x^2 + 6x - 7 = (x + 7)(x - 1)$$

So:

$$\frac{(x + 7)(x - 1)}{x + 7} = x - 1 \quad \text{for } x \neq -7$$

Thus,

$$ax + d = x - 1 \Rightarrow a = 1, \quad d = -1$$

Therefore,

$$a + d = 1 + (-1) = 0$$

Correct Answer: C

Question: Simplifying Rational Expressions

Let:

$$f(x) = x^3 - 9x, \quad g(x) = x^2 - 2x - 3$$

Which of the following expressions is equivalent to:

$$\frac{f(x)}{g(x)} \quad \text{for } x > 3$$

Options:

A. $\frac{1}{x+1}$

B. $\frac{x+3}{x+1}$

C. $\frac{x(x-3)}{x+1}$

D. $\frac{x(x+3)}{x+1}$

Answer Explanation: Simplifying Rational Expressions

We start by factoring both the numerator and denominator:

$$f(x) = x^3 - 9x = x(x^2 - 9) = x(x - 3)(x + 3)$$

$$g(x) = x^2 - 2x - 3 = (x - 3)(x + 1)$$

Now simplify:

$$\frac{f(x)}{g(x)} = \frac{x(x - 3)(x + 3)}{(x - 3)(x + 1)} = \frac{x(x + 3)}{x + 1}$$

Correct Answer: D

Fraction

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Adding Algebraic Fractions

To add or subtract algebraic fractions:

- Step 1: Find a common denominator.
- Step 2: Rewrite each fraction with the common denominator.
- Step 3: Combine the numerators.
- Step 4: Simplify if possible.

Example:

$$\frac{1}{x} + \frac{1}{x+1}$$

Step 1: Common denominator is $x(x+1)$

$$\frac{1}{x} = \frac{x+1}{x(x+1)}, \quad \frac{1}{x+1} = \frac{x}{x(x+1)}$$

Step 2: Add the numerators:

$$\frac{x+1}{x(x+1)} + \frac{x}{x(x+1)} = \frac{(x+1) + x}{x(x+1)} = \frac{2x+1}{x(x+1)}$$

Tip: Always check if you can factor or simplify the final expression.

Flipping and Splitting Fractions

1. Flipping Fractions (Reciprocals):

Rule: When dividing by a fraction, multiply by its reciprocal.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

Example:

$$\frac{3}{4} \div \frac{2}{5} = \frac{3}{4} \cdot \frac{5}{2} = \frac{15}{8}$$

Tip: Always flip the **second** fraction and multiply.

2. Splitting Fractions:

Rule: A fraction with addition or subtraction in the numerator can be split into two separate fractions:

$$\frac{a + b}{c} = \frac{a}{c} + \frac{b}{c}, \quad \frac{a - b}{c} = \frac{a}{c} - \frac{b}{c}$$

But: You **cannot** split a denominator:

Question: Adding Rational Expressions

$$\frac{2}{x-2} + \frac{3}{x+5} = \frac{rx+t}{(x-2)(x+5)}$$

This equation is true for all $x > 2$, where r and t are positive constants. What is the value of rt ?

Options:

- A. -20
- B. 15
- C. 20
- D. 60

Answer Explanation: Adding Rational Expressions

Start by finding a common denominator:

$$\frac{2}{x-2} + \frac{3}{x+5} = \frac{2(x+5) + 3(x-2)}{(x-2)(x+5)}$$

Now expand the numerator:

$$2(x+5) + 3(x-2) = 2x + 10 + 3x - 6 = 5x + 4$$

So the left-hand side becomes:

$$\frac{5x + 4}{(x-2)(x+5)}$$

Compare with the right-hand side:

$$\frac{rx + t}{(x-2)(x+5)} \Rightarrow rx + t = 5x + 4$$

So:

Isolate a Variable

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Cross-Multiplication Technique

When you have an equation of the form:

$$\frac{A}{B} = \frac{C}{D}$$

You can use **cross-multiplication** to eliminate denominators:

$$A \cdot D = B \cdot C$$

Example:

$$\frac{u-3}{1} = \frac{6}{t-2} \Rightarrow (u-3)(t-2) = 6$$

Why it works: This is based on the property of proportions. Multiplying both sides by the least common denominator (LCD) clears all denominators in one step.

Clearing Square Roots: Squaring Both Sides

If a variable is under a square root, square both sides to eliminate the radical.

General Rule:

$$\sqrt{E} = F \Rightarrow E = F^2$$

Example:

$$2\sqrt{w+19} = \frac{14x}{7y} \Rightarrow \sqrt{w+19} = \frac{2x}{y} \Rightarrow w+19 = \left(\frac{2x}{y}\right)^2$$

Tip: Always isolate the radical before squaring both sides.

Solving Literal Equations (Isolate a Variable)

Literal equations are formulas or equations involving multiple variables. The goal is to **solve for one variable in terms of the others**.

Key Steps:

- Step 1: Identify the variable you want to isolate (make it the subject).
- Step 2: Use inverse operations (addition, subtraction, multiplication, division) to isolate that variable.
- Step 3: If needed, eliminate radicals or simplify fractions.

Tips:

- Use cross-multiplication when variables appear in denominators.
- Clear square roots by squaring both sides of the equation.
- Watch for extraneous solutions if squaring or multiplying both sides.

Question: Solving for w in terms of x and y

Given the equation:

$$\frac{14x}{7y} = 2\sqrt{w + 19}$$

The equation relates the distinct positive real numbers w , x , and y . Which equation correctly expresses w in terms of x and y ?

Options:

A. $w = \sqrt{\frac{x}{y}} - 19$

B. $w = \sqrt{\frac{28x}{14y}} - 19$

C. $w = \left(\frac{x}{y}\right)^2 - 19$

D. $w = \left(\frac{28x}{14y}\right)^2 - 19$

Answer Explanation: Solving for w in terms of x and y

Start with the equation:

$$\frac{14x}{7y} = 2\sqrt{w+19} \Rightarrow 2\sqrt{w+19} = 2\sqrt{w+19} \Rightarrow \frac{14x}{7y} = 2\sqrt{w+19} \Rightarrow \frac{2x}{y} =$$

Square both sides:

$$\left(\frac{2x}{y}\right)^2 = w + 19 \Rightarrow \frac{4x^2}{y^2} = w + 19 \Rightarrow w = \frac{4x^2}{y^2} - 19$$

Now compare this with the answer choices. Only option **C** correctly represents the structure:

$$w = \left(\frac{x}{y}\right)^2 - 19$$

Correct Answer: C

Question: Solving for t in terms of u

Given the equation:

$$u - 3 = \frac{6}{t - 2}$$

What is t in terms of u ?

Options:

- A. $t = \frac{1}{u}$
- B. $t = \frac{2u+9}{u}$
- C. $t = \frac{1}{u-3}$
- D. $t = \frac{2u}{u-3}$

Answer Explanation: Solving for t in terms of u

Starting from the equation:

$$u - 3 = \frac{6}{t - 2}$$

Multiply both sides by $(t - 2)$:

$$(u - 3)(t - 2) = 6$$

Expand the left-hand side:

$$ut - 2u - 3t + 6 = 6 \Rightarrow ut - 3t = 2u \Rightarrow t(u - 3) = 2u \Rightarrow t = \frac{2u}{u - 3}$$

Correct Answer: D

Practice

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Question: Adding Algebraic Fractions

Which expression is equivalent to

$$\frac{y - 3xy}{2x^2 - 2xy} + \frac{1 + 3y}{x - y} ?$$

Options:

A. $\frac{2x - 3xy + 4y}{2x^2 - 2xy}$

B. $\frac{2x + 3xy + y}{2x^2 - 2xy}$

C. $\frac{2x + 3xy + y}{2x^3 - 4x^2y + 2xy^2}$

D. $\frac{4y - 3xy + 1}{2x^3 - 4x^2y + 2xy^2}$

Answer Explanation: Adding Algebraic Fractions

We simplify each term separately:

First term:

$$\frac{y - 3xy}{2x^2 - 2xy} = \frac{y(1 - 3x)}{2x(x - y)}$$

Second term:

$$\frac{1 + 3y}{x - y}$$

To combine these, we need a common denominator. The least common denominator (LCD) is $2x(x - y)$.

Rewrite the second term using LCD:

$$\frac{1 + 3y}{x - y} = \frac{(1 + 3y) \cdot 2x}{2x(x - y)} = \frac{2x + 6xy}{2x(x - y)}$$

Now add the two terms:

$$\frac{y - 3xy + 2x + 6xy}{2x^2 - 2xy} = \frac{2x + 3xy + y}{2x^2 - 2xy}$$

Question: Solving for k in terms of x

If $\frac{4}{k+2} = \frac{x}{3}$, where $k \neq -2$, which equation correctly expresses k in terms of x ?

Options:

A. $k = \frac{12 - 2x}{x}$

B. $k = \frac{12 + 2x}{x}$

C. $k = \frac{x}{12 + 2x}$

D. $k = 12x - 2$

Answer Explanation: Solving for k in terms of x

We are given:

$$\frac{4}{k+2} = \frac{x}{3}$$

Cross-multiply:

$$4 \cdot 3 = x(k+2) \Rightarrow 12 = xk + 2x$$

Solve for k :

$$12 - 2x = xk \Rightarrow k = \frac{12 - 2x}{x}$$

Correct Answer: A

Question: Solving for G in terms of L , P , and T

$$T = 2P\sqrt{\frac{L}{G}}$$

The given equation relates the positive real numbers G , L , P , and T . Which equation correctly expresses G in terms of L , P , and T ?

Options:

- A. $\frac{2PL}{T^2}$
- B. $\frac{T^2}{4LP^2}$
- C. $\frac{LT^2}{4P^2}$
- D. $\frac{4LP^2}{T^2}$

Answer Explanation: Solving for G in terms of L , P , and T

We start with the equation:

$$T = 2P\sqrt{\frac{L}{G}}$$

Divide both sides by $2P$:

$$\frac{T}{2P} = \sqrt{\frac{L}{G}}$$

Square both sides:

$$\left(\frac{T}{2P}\right)^2 = \frac{L}{G} \Rightarrow \frac{T^2}{4P^2} = \frac{L}{G}$$

Solve for G :

$$G = \frac{4P^2L}{T^2}$$

Correct Answer: D

Question: Integer Relationship in Factored Form

The expression

$$4x^2 + bx - 45$$

where b is a constant, can be rewritten as

$$(hx + k)(x + j),$$

where h , k , and j are **integer constants**. Which of the following must be an integer?

Options:

- A. $\frac{b}{h}$
- B. $\frac{b}{k}$
- C. $\frac{45}{h}$
- D. $\frac{45}{k}$

Answer Explanation: Integer Relationship in Factored Form

We are given:

$$4x^2 + bx - 45 = (hx + k)(x + j)$$

Expanding the right-hand side:

$$hx^2 + (hj + k)x + kj$$

By comparing coefficients:

$$h = 4, \quad b = hj + k, \quad kj = -45$$

Since k and j are integers and their product $kj = -45$, it follows that k must divide 45. Therefore:

$$\frac{45}{k} \text{ must be an integer.}$$

Correct Answer: D

Thank You!

We appreciate your time and focus.

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